Unit 1 Summary: Displaying One Variable

Name _____

Showing the Data

Display	How it works	Example
Frequency table	Lists each value and its **frequency** (how many times it happens). Make a tally mark for each value. Then count the marks.	11 students say how many siblings they have. 4 of them have 1 sibling. Check: $2 + 4 + 3 + 1 + 1 = 11$ students.
Dot plot	Puts the values on a number line. Each value gets **one dot**. Dots for the same value stack up. The height of a stack is the frequency. It shows every value.	The same siblings data. The stack above 1 is the tallest, so 1 is the most common.
Histogram	A bar graph for number data. The values are put into groups called **bins**. Each bar is one bin. The bars touch (no gaps). It does **not** show the exact values. It is better than a dot plot when there is a lot of data, or the values are spread out.	Heights of students. The tallest bar is 170 to 180. So 7 students are from 170 cm to 180 cm tall.

The histogram rule: if a value is **on a line** between two bars, it goes in the bar on the **right**. 160 goes in the 160 to 170 bar.

Every time you read a graph: read the title, read both axis labels, then check the scale.

Center

Measure	How to find it	Example: 2, 3, 8, 3, 4
Mean (the average)	Add all the values. Divide by how many values there are. In symbols: $\bar{x} = \frac{\Sigma x}{n}$. Σ (sigma) means **add up**. n is how many values there are. $\bar{x}$ ("x-bar") is the mean.	$\Sigma x = 2 + 3 + 8 + 3 + 4 = 20$ and $n = 5$, so $\bar{x} = \frac{20}{5} = 4$.
Median (the middle)	Put the values in order. Find the middle value. Two values in the middle? The median is the number halfway between them.	In order: 2, 3, **3**, 4, 8. The median is 3. For 2, 3, 5, 8: $(3 + 5) \div 2 = 4$.
Mode	The value that happens **most often**.	The mode is 3.

Shape

Shape	What it looks like	Mean and median
Skewed left	Tail on the left	mean $<$ median
Symmetric	Both sides match	mean $\approx$ median
Skewed right	Tail on the right	mean $>$ median

The mean follows the tail. The tail is the long, low side of a graph. The mean is pulled toward the tail. The median is not pulled much. With a long tail, the median is a better description of a **typical** value.

Quartiles and Spread

How to find it	Example 1: 11 values (odd)	Example 2: 10 values (even)
1. Put the data in order.	3, 4, 6, 7, 9, **10**, 12, 13, 15, 18, 20	1, 3, 4, 6, **7**, **9**, 10, 12, 14, 18
2. Find the **median**. It splits the data into a lower half and an upper half.	The median is 10. **Odd number of values:** leave the median out of both halves.	Two middle values, 7 and 9. The median is $(7 + 9) \div 2 = 8$.
3. **Q1** = the median of the lower half.	Lower half: 3, 4, **6**, 7, 9. Q1 = 6	Lower half: 1, 3, **4**, 6, 7. Q1 = 4
4. **Q3** = the median of the upper half.	Upper half: 12, 13, **15**, 18, 20. Q3 = 15	Upper half: 9, 10, **12**, 14, 18. Q3 = 12
Five-number summary: minimum, Q1, median, Q3, maximum	3, 6, 10, 15, 20	1, 4, 8, 12, 18
IQR = $Q_3 - Q_1$	$15 - 6 = 9$	$12 - 4 = 8$

IQR (interquartile range): how spread out the **middle half** of the data is. **Range** = maximum − minimum. One extreme value changes the range a lot. It does not move the IQR much.

Box Plots

Part	What it shows	Example: test scores
The box	Goes from Q1 to Q3. Its length is the IQR. It holds the middle **50%** of the data.	Half of the students scored from 68 to 84.
The line inside the box	The median. About **50%** of the data is below it.	Half scored less than 76.
The whiskers	Go out to the minimum and the maximum.	From 52 to 68, and from 84 to 98.
Each of the 4 parts	About **25%** of the data. A long part means the values are **spread out**. It does not mean there are more of them.	The left whisker (52 to 68) is the longest part. It still holds only about 25% of the students.
Above Q1	About **75%** of the data.	About 75% scored more than 68.

Compare two box plots: put both on the same number line. Then compare the centers and the spreads.

Compare	Morning	Afternoon	What it means
Center (median)	72	78	The afternoon class scored **higher**.
Spread (IQR)	$80 - 65 = 15$	$82 - 70 = 12$	The afternoon scores are **closer together**.

Outliers

The 1.5 × IQR rule	Example: 3, 8, 9, 10, 11, 12, 13, 14, 30
1. Find Q1, Q3, and the IQR.	Q1 = 8.5, Q3 = 13.5, and IQR = 5.
2. Lower fence = $Q_1 - 1.5 \times$ IQR	$1.5 \times 5 = 7.5$. Lower fence = $8.5 - 7.5 = 1$
3. Upper fence = $Q_3 + 1.5 \times$ IQR	Upper fence = $13.5 + 7.5 = 21$
4. A value below the lower fence or above the upper fence is an **outlier**. It is far away from the rest of the data.	30 is above 21, so **30 is an outlier**. 3 is not below 1, so 3 is not an outlier.

On a box plot: an outlier is a **dot**. The whisker stops at the last value that is not an outlier (here, 14).